

Recording 2: Yaricsa and Tatitana (2)

Researcher 00:01

First, thank you so much for meeting with me. And are we interested here in body experience of using aladdin to design the that yellow house? There are no right or wrong answers. I just want to understand your process, so it's no right or wrong answers. And you can just be honest, it's awful, it's fine. And I have some opening questions. I think I asked in the class, but could you share again, like, which like journal ai tools have you used before this project?

Student1 00:41

I've used google gemini that's about the only one. I can't really do check gpt with others.

Researcher 00:47

I use that too.

Student2 00:49

I've only ever used gemini ai gemini, that's the only thing i've ever had.

Researcher 00:54

I think it's mainly because you are with this google system, right? I use that too. My next question is that what was your first impression of using ai for a complex engineering task like this?

Student1 01:12

At first, I was kind of confused, but after a few prompts, I got used to it. It was really easy to use after the first I.

Researcher 01:21

see.

Student2 01:22

same thing here, but I thought it was really cool that you were able to just build a house technically using ai making it.

Student1 01:35

easier.

Researcher 01:36

Thank you. And again, you don't always say good things, just be honest

because it's a way for us to improve. He can say it is terrible. I don't believe this one. I hope this is working this way. It's terrifying. Okay? So my next questions, I I showed you the three primary questions, and let's start with the first one. My first question is that were new science and engineering concepts. Have you learned from using the tool that we developed? Like anything that you can think about science on your engineering,

Student2 02:11

the lights, like where the sun hits, the house and like the mountain windows you used. Yeah.

The sunlight going into the house and the heat and stuff like that. I.

Researcher 02:25

see what about you.

Student1 02:27

that you have to take insulation into consideration to like help save energy. What is.

Researcher 02:34

insulation? By the way,

Student2 02:36

the stuff that goes like in the walls of the house.

Student1 02:39

like helps keep the house cool or warm.

Researcher 02:43

I see good everything. Do you know this before ago? You learned through the process? I.

Student2 02:54

already knew some stuff before.

Student1 02:56

and about science, but not much about engineering. I.

Researcher 03:00

see. I see, did you notice that ai reasoning, when you use a tool, there's aai reasoning part. Have you read that part?

Student1 03:11

I've read it.

Researcher 03:12

What do you notice? Did they provide any important information? Or is it useless? Most of the time?

Student1 03:18

I noticed how it took, what I wrote, and then it was like like a response, but it told me the changes it made to the house,

Researcher 03:26

made the changes.

Student1 03:29

Yeah, and how it added and removed the stuff I wanted to,

Researcher 03:32

not much that science focus, right?

Student2 03:36

Not really. Now,

Researcher 03:37

that's what my impression, too, is my impression, too. By any chance you see that ai ai reasoning mention anything that you feel that helps to think about the design of science.

Student2 03:55

I feel like you should give like a couple examples, example. People could get an idea of how to write a prompt for life, just how to use it in general, and how it would change like the building of the house.

Researcher 04:10

Can you give me more specific example?

Student2 04:14

Like when a person doesn't know how to write a prompt or how to properly write it? They could just use an example from there and get ideas.

Student1 04:27

Think.

Researcher 04:29

about how to do like how to be specific about the design, not that much about the science. I think that is something we will have to address.

The next question that I showed you is a what strategies you learn from ai in designing towards a net zero goal, like a a like when, for example, when you design like any strategy you used?

Student1 05:01

Well, like a strategy, I like what we did to make it,

I know I did, I increase the r value, decrease the u value, because they are inverse. They have an inverse relationship. When one is high, other is low, which is which balances out makes them both good.

Researcher 05:24

What about you?

Student2 05:25

Basically, the.

Researcher 05:27

same thing I see, did you realize the air reasoning has something about in good information about the u and r value in the air reasoning or? No, thank you. I did. What do you say? I.

Student1 05:41

I forgot to lose it. It was a nice paragraph. I skimmed over it.

Researcher 05:47

I see, it's very lengthy, right? I think.

Student1 05:50

it's good. It's detail the.

Researcher 05:52

details. I see what kind of information you get from. That are reasoning part about the details, about these values.

Student1 06:00

It did mention like about the east, west for the solar panels.

a little bit. I didn't know about that before. So.

Researcher 06:10

it's like the look the facing the direction of the solar panel matters. I see, it does provide some information about the design right direction for the design. Yeah, interesting.

And then the third question I showed you is.

my first next question is that what's your opinion about the pros and cons of using ai in engineering design?

Student2 06:47

Sometimes it's not gonna be as accurate as ai would be, because ai is just a computer, so it's not actually there to see the front.

Student1 07:00

But ii do think it's really good for like getting an idea of the model you want and then go on from there.

Researcher 07:09

Yeah. So let's think about the problems. Think about like in what ways did I make you like a better, faster designer? Did you help you visualize things you couldn't before like the ai like using prompt to generate buildings?

Student2 07:27

It really did a.

Researcher 07:29

fast designer,

Student2 07:30

a clear image of like what I wanted.

Researcher 07:33

I see. Yeah. And about the columns, we all know has hallucination, right? Did you like ever feel like that? Ai was hallucinate in or give you technically impossible suggestions? May I that I don't think so right? In this case, I don't think so, because it's during the building. Right?

Yeah. I do see that students in the class. They don't realize the prompt. They just click the walls, the solar panels, and then do that manually without the prompt. It seems that it should be by bidirectional. What you're feeling about that because I see, but I demo in the ways of use prompted to generate and revise the design. But when I look around, many students don't use prompt, they drag the walls and the drag the solar panel from the bottoms to add solar panels. What do you think about in this? Like using texture design? Whether is you more hands on, you click buttons to do it. What do you see the pros and cons about these actions?

Student2 08:47

I think it's really good because there's students that really just wanna like do it themselves and build it themselves and learn along the way while doing it.

Student1 08:55

I.

Researcher 08:56

see.

Student1 08:58

they could like make some measurement errors, which I think that's where ai could really help, because ai is good measurement, but they could what take the measurements and then do it on their own.

Researcher 09:13

I I see also prompt one thing about prompt, ii think, is very helpful is that you can be free style in one class. One student share that his strategy over cutting the energy cost is say that you may, I don't need ac even though you put ac there, but I don't need ac with amaze. I can bear with that kind of heat, you may. And then that kind of energy costs. So this is impossible to do it a with a jacket, the job that tool, right? You specify that like a specific requirements, details, impromptu. So the way I can do it could be it's tied back to your to idea about it is a very precise. Thank you. This question about like at any point, did you feel like ai was doing the thinking for you. How do you ensure the final design was still yours? Is not like a is. It's not because it's true based on prompt,

Student1 10:18

right? It's based off I what I put in. Do.

Researcher 10:22

you feel that? Do you thinking for you? No. In this case, right?

Have you used ai tools that you feel is doing thinking for you? And you don't feel that's yours after you use it?

Student2 10:36

I feel like if i'm not being specific enough, maybe it can be thinking for itself.

Researcher 10:42

I see. I I see that makes sense. My next question is about future. If you were a professional engineering tomorrow, maybe not tomorrow in a few years, would you want this tool by your side? Why or why not? I.

Student2 10:58

would.

Student1 10:59

I think it would be a really useful tool to like help me build, even because my brother he plans on going to engineer design, I really think it could be helpful for him, too, because he's more of a visual person in ai it does the visually. And he could go from there.

Student2 11:19

Yes, I really like the visual idea. I like the thought of it being like, you're a person being able to see the design you've created and and come to life.

Researcher 11:31

What about the part about the use, like prompt to design, to do the design, like you use voice or use text to turn designs?

Student2 11:44

I think that's good, too for the people that think of an idea, but don't know how to put the idea in the design. I think you.

Researcher 11:55

can be awake in prom, right? So I have a final question is that is there anything else about experience without aladdin or net zero house design that ii haven't asked anything else you want to share?

Student1 12:15

Yeah, it was good. That I didn't have any problems with it. There was no like anything that it did. It was good.

Student2 12:23

I feel like it needs like at least a mini tutorial to like. So I can guide myself through creating the design or the house. I.

Researcher 12:34

I see what kind of tutorial I thinking about,

Student2 12:36

like, just like using different tools and being hands on and how to properly use the prompt.

Researcher 12:43

I see, i'll type it in some like example or prompts. I think that's a good idea. I would include that too. Yeah, cool. So I don't have other questions. Thank you very much. It's very short, only 12 minutes, which is.